SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 35%

Dr.T.P.Anithaashri

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Industry Problem Task **Duration: 2.5hrs**

1. Explain how Reinforcement Learning can be applied to optimize delivery routes in a logistics system. Identify the possible states, actions, rewards, and policy. Discuss how continuous learning improves efficiency over time. **(5 Marks)**

2. Apply Reinforcement Learning concepts to a fraud detection system in fintech. Define the state space, action space, and reward mechanism. Explain how exploration strategies improve detection accuracy. **(5 Marks)**

3. Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance. **(5 Marks)**

4. Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments. **(5 Marks)**

5. Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance. **(5 Marks)**

6. Apply Reinforcement Learning to dynamic pricing in e-commerce. Define the state space, action space, and reward function. Discuss challenges in balancing exploration and exploitation. **(5 Marks)**

7. Illustrate the use of Reinforcement Learning in smart city waste management systems. Identify the MDP components and discuss how continuous updates improve operational efficiency. **(5 Marks)**

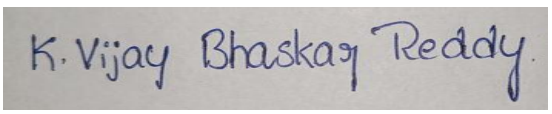
8. Evaluate how Reinforcement Learning can be used in network bandwidth allocation in telecommunications. Define states, actions, and rewards, and examine how the system adapts to dynamic conditions. **(5 Marks)**

9. Analyze the application of Reinforcement Learning in portfolio management in finance. Discuss advantages over traditional models and challenges such as risk and delayed rewards. **(5 Marks)**

10. Justify the use of Reinforcement Learning in renewable energy management systems. Define the RL framework components and discuss the impact of environmental uncertainty on decision-making. **(5 Marks)**

Code of Conduct:

I K.Vijay Bhaskar Reddy(192425155) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. Explain how Reinforcement Learning can be applied to optimize delivery routes in a logistics system. Identify the possible states, actions, rewards, and policy. Discuss how continuous learning improves efficiency over time.

Aim

To optimize delivery routes using Reinforcement Learning (Q-Learning).

Theory

Reinforcement Learning enables a delivery agent to learn the shortest path to a destination by interacting with the environment. The agent receives rewards for reaching the destination and penalties for taking unnecessary steps or hitting obstacles. Over multiple episodes, the Q-table is updated, allowing the agent to discover the optimal route.

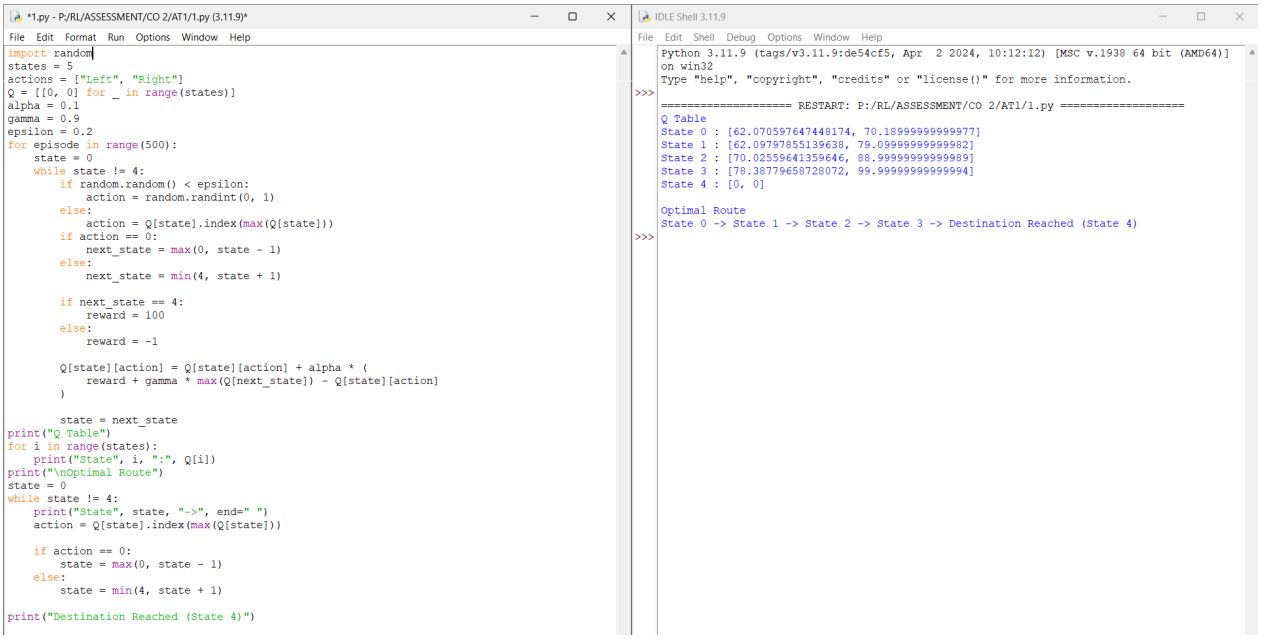
MDP Components

- **States:** Grid locations (0–15)
- **Actions:** Up, Down, Left, Right
- **Reward:** +100 (Destination), -1 (Move), -100 (Obstacle)
- **Policy:** Choose the action with the highest Q-value

Algorithm

Q-Learning

Python Code



Code Explanation

Step 1

Create **5 states** representing delivery locations.

states = 5

Step 2

Initialize the **Q-table** with zeros.

Q = [[0,0] for _ in range(states)]

Step 3

Train the agent for **500 episodes**.

for episode in range(500):

Step 4

Choose an action using the **ε-greedy strategy**.

- Explore randomly.
- Otherwise, choose the best known action.

Step 5

Assign rewards.

- **+100** when the destination is reached.
- **-1** for every normal move.

Step 6

Update the Q-table using the **Q-Learning formula**:

Q(s,a)=Q(s,a)+α[r+γmaxQ(s′,a′)−Q(s,a)]Q(s,a)=Q(s,a)+\alpha\left[r+\gamma\max Q(s',a')-Q(s,a)\right]Q(s,a)=Q(s,a)+α[r+γmaxQ(s′,a′)−Q(s,a)]

where:

- **α (alpha)** = Learning Rate
- **γ (gamma)** = Discount Factor
- **r** = Reward

Step 7

After training, the agent follows the highest Q-value at each state to reach the destination.

Advantages

- Learns automatically from experience.
- Adapts to traffic changes.
- Reduces delivery time.

Applications

- Amazon
- Flipkart
- Swiggy
- Zomato
- Warehouse robots

Conclusion

Reinforcement Learning helps logistics companies reduce travel time and improve delivery efficiency by continuously learning better routes.

2. Apply Reinforcement Learning concepts to a fraud detection system in fintech. Define the state space, action space, and reward mechanism. Explain how exploration strategies improve detection accuracy.

Introduction

Financial fraud causes significant losses to banks and digital payment systems. Reinforcement Learning helps detect fraud by learning transaction patterns and adapting to new fraud techniques.

RL Components

State Space

The state includes information about each transaction.

- Transaction amount
- User location
- Device information
- Time of transaction
- Account balance
- Transaction history

Action Space

The agent can:

- Approve transaction
- Reject transaction
- Flag transaction for manual review

Reward Mechanism

- +10 for detecting fraud
- +5 for approving legitimate transactions
- -15 for approving fraudulent transactions
- -10 for blocking genuine customers

Policy

The policy selects the action with the highest expected reward based on transaction features.

Exploration Strategy

Initially, the agent explores different detection strategies. Over time, it exploits successful detection patterns while occasionally exploring new behaviors to detect emerging fraud techniques.

Advantages

- Detects unknown fraud patterns
- Reduces financial loss
- Learns continuously
- Improves detection accuracy
- Reduces false positives

Challenges

- Delayed feedback**Aim**

  To implement a simple Reinforcement Learning (Q-Learning) algorithm for fraud detection by learning whether to **Approve** or **Block** a transaction

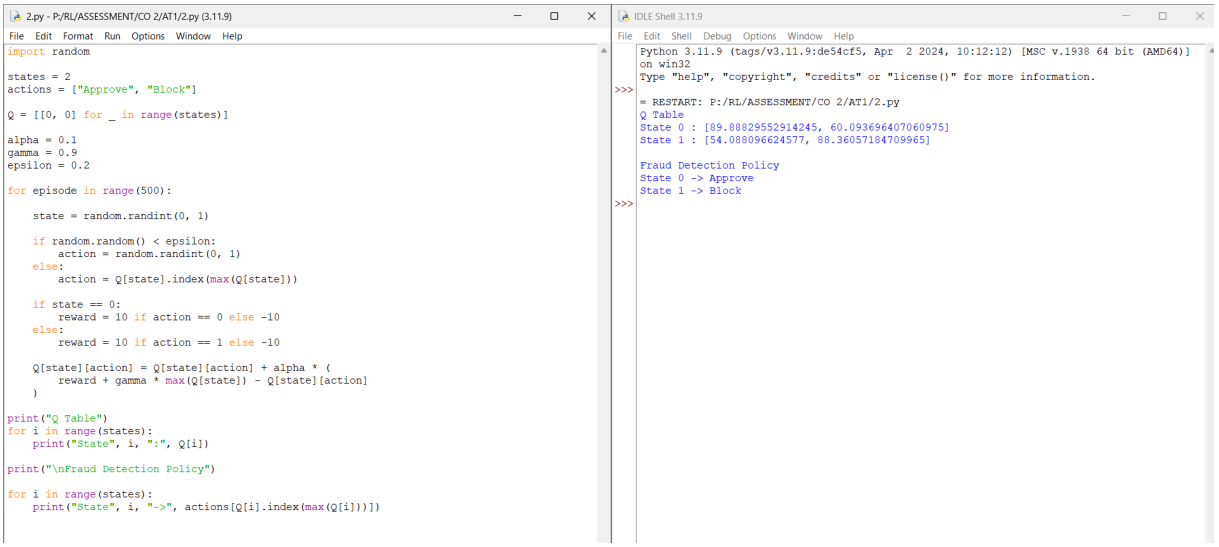
Theory

Reinforcement Learning (RL) enables a fraud detection system to learn the best decision by interacting with transaction data. The agent receives a **reward** for correctly blocking fraudulent transactions and a **penalty** for approving fraud. Over time, it learns the best action for each transaction state.

MDP Components

Component	Description
States	Genuine Transaction, Suspicious Transaction
Actions	Approve, Block
Reward	+10 for correct decision, -10 for wrong decision
Policy	Choose the action with the highest Q-value

Python Code



Code Explanation

Step 1

Create **2 transaction states**:

- State 0 → Genuine Transaction
- State 1 → Fraudulent Transaction

states = 2

Step 2

Define the possible actions.

actions = ["Approve", "Block"]

Step 3

Initialize the Q-table with zeros.

Q = [[0,0] for _ in range(states)]

Step 4

Train the agent for **500 episodes**.

for episode in range(500):

Step 5

Choose an action using the **ε-greedy policy**.

- Explore randomly.
- Otherwise, choose the best known action.

Step 6

Assign rewards.

- Genuine transaction approved → **+10**
- Fraud blocked → **+10**
- Wrong decision → **-10**

Step 7

Update the Q-table using the Q-Learning formula.

After training, the agent learns:

- **Approve** genuine transactions.
- **Block** fraudulent transactions.

Advantages

- Learns automatically from transaction history.
- Improves fraud detection accuracy.
- Reduces financial losses.
- Adapts to new fraud patterns.

Applications

- Online Banking
- Credit Card Fraud Detection
- UPI Payment Security
- E-Commerce Payment Systems
- Digital Wallets

Conclusion

This program demonstrates how **Q-Learning** can be applied to fraud detection. By receiving rewards for correct decisions and penalties for incorrect ones, the agent learns an effective policy to approve genuine transactions and block fraudulent ones, improving detection accuracy over time.

3. Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance.

Aim

To implement a simple Reinforcement Learning (Q-Learning) algorithm for recommending content based on user feedback.

Theory

Reinforcement Learning (RL) helps streaming platforms recommend content by learning from user interactions. The agent receives positive rewards when users like the recommended content and negative rewards when users skip it. Over time, it learns the best recommendation for each user.

MDP Components

Component	Description
States	Movie Lover, Music Lover
Actions	Recommend Movie, Recommend Music
Rewards	+10 for correct recommendation, -5 for incorrect recommendation
Policy	Select the action with the highest Q-value

Program

Code Explanation

Step 1: Define two user preference states.

- State 0 → Movie Lover
- State 1 → Music Lover

Step 2: Define two actions.

- Recommend Movie
- Recommend Music

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Choose an action using the ε-greedy strategy.

Step 6: Give rewards.

- Correct recommendation → +10
- Incorrect recommendation → -5

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Print the learned recommendation policy.

Advantages

- Learns user preferences automatically.
- Improves recommendation accuracy.

- Adapts to changing user interests.
- Increases user engagement.

Applications

- Netflix
- YouTube
- Spotify
- Amazon Prime Video
- Disney+ Hotstar

Conclusion

This program demonstrates how Reinforcement Learning can recommend personalized content by learning from user feedback. The Q-table improves over time, enabling the system to recommend content that maximizes user satisfaction.

Q4. Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments. (5 Marks)

Aim

To evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches.

Theory

Reinforcement Learning (RL) enables machines to learn the best maintenance schedule by continuously monitoring equipment conditions. Unlike rule-based systems, RL adapts to changing machine behavior and predicts failures before they occur, reducing downtime and maintenance costs.

MDP Components

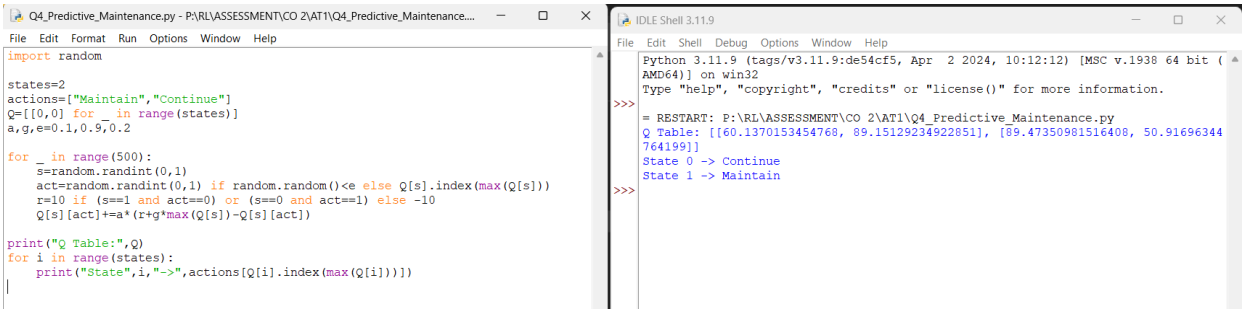
States: Machine Healthy, Machine Faulty

Actions: Perform Maintenance, Continue Operation

Rewards: +10 for timely maintenance, -10 for machine failure

Policy: Select the action with the highest Q-value.

Python Code



Code Explanation

Step 1: Define two machine states.

- State 0 → Healthy Machine
- State 1 → Faulty Machine

Step 2: Define two actions.

- Perform Maintenance
- Continue Operation

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Select an action using the ε-greedy strategy.

Step 6: Give rewards.

- Correct maintenance decision → **+10**
- Incorrect decision → **-10**

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Display the best maintenance policy.

Advantages

- Reduces equipment failures.
- Minimizes maintenance costs.
- Improves machine reliability.
- Learns automatically from experience.

Applications

- Manufacturing industries
- Smart factories
- Power plants
- Industrial robots

Conclusion

Reinforcement Learning improves predictive maintenance by learning optimal maintenance schedules, reducing downtime, and increasing equipment reliability

Q5. Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance. (5 Marks)

Aim

To optimize bus scheduling and route allocation using Reinforcement Learning.

Theory

Reinforcement Learning helps transportation systems improve bus scheduling by analyzing passenger demand and traffic conditions. The agent learns to allocate buses efficiently, reducing waiting time and improving service quality.

MDP Components

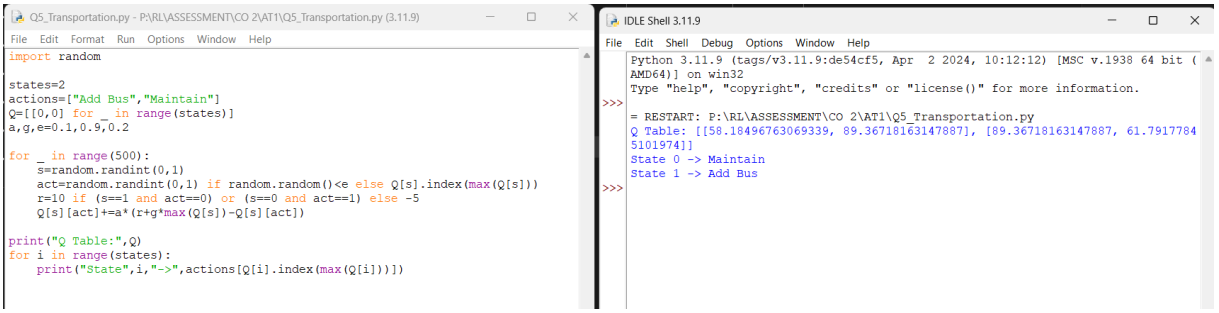
States: Low Traffic, High Traffic

Actions: Add Bus, Maintain Schedule

Rewards: +10 for efficient scheduling, -5 for delays

Policy: Select the best scheduling action.

Python Code



Code Explanation

Step 1: Define two traffic states.

- State 0 → Low Traffic
- State 1 → High Traffic

Step 2: Define two actions.

- Add Bus
- Maintain Schedule

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Select an action using the ε-greedy strategy.

Step 6: Give rewards.

- Correct scheduling → **+10**
- Incorrect scheduling → **-5**

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Display the best bus scheduling policy.

Advantages

- Reduces passenger waiting time.
- Optimizes bus allocation.
- Adapts to traffic changes.
- Improves transport efficiency.

Applications

- Public transportation
- Metro systems
- Smart traffic management
- Logistics routing

Conclusion

RL enables transportation systems to make intelligent scheduling decisions using real-time traffic and passenger information.

Q6. Apply Reinforcement Learning to dynamic pricing in e-commerce. Define the state space, action space, and reward function. Discuss challenges in balancing exploration and exploitation.

Aim

To apply Reinforcement Learning for optimizing product prices based on demand.

Theory

RL learns the best pricing strategy by adjusting prices according to customer demand and sales performance. It balances exploration and exploitation to maximize long-term profit.

MDP Components

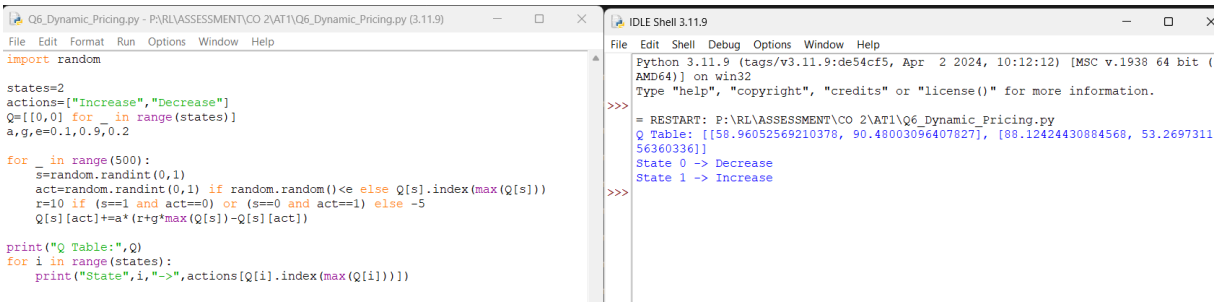
States: Low Demand, High Demand

Actions: Increase Price, Decrease Price

Rewards: +10 for higher profit, -5 for lower sales

Policy: Select the price with the highest Q-value.

Python Code



Code Explanation

Step 1: Define two demand states.

- State 0 → Low Demand
- State 1 → High Demand

Step 2: Define two actions.

- Increase Price

- Decrease Price

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Select an action using the ε-greedy strategy.

Step 6: Give rewards.

- Correct pricing decision → **+10**
- Incorrect pricing decision → **-5**

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Display the optimal pricing policy.

Advantages

- Maximizes profit.
- Adapts to market demand.
- Learns pricing strategies automatically.
- Improves customer satisfaction.

Applications

- Amazon
- Flipkart
- Airline ticket pricing
- Hotel booking systems

Conclusion

RL enables businesses to determine optimal product prices while balancing revenue and customer demand.

Q7. Illustrate the use of Reinforcement Learning in smart city waste management systems. Identify the MDP components and discuss how continuous updates improve operational efficiency. (5 Marks)

Aim

To optimize waste collection using Reinforcement Learning.

Theory

RL helps waste collection vehicles choose efficient routes based on the fill level of garbage bins. Continuous sensor updates enable timely collection and reduce operational costs.

MDP Components

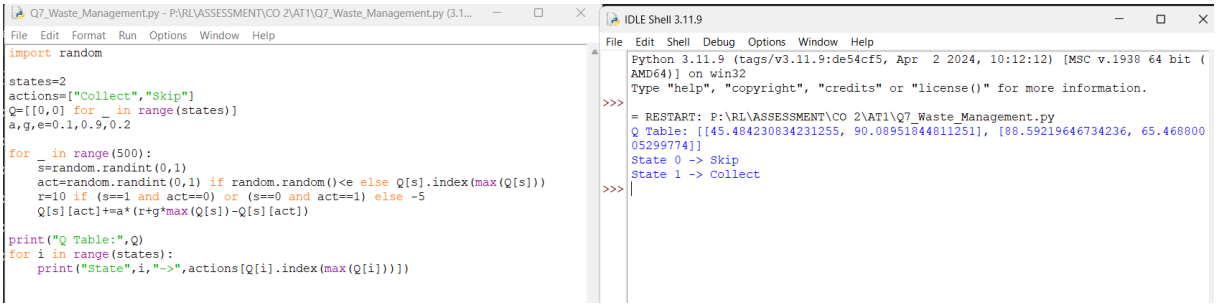
States: Bin Empty, Bin Full

Actions: Collect Waste, Skip Bin

Rewards: +10 for timely collection, -5 for overflow

Policy: Select the best collection action.

Python Code



Code Explanation

Step 1: Define two bin states.

- State 0 → Bin Empty
- State 1 → Bin Full

Step 2: Define two actions.

- Collect Waste
- Skip Bin

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Select an action using the ε-greedy strategy.

Step 6: Give rewards.

- Timely collection → **+10**
- Wrong decision → **-5**

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Display the best waste collection policy.

Advantages

- Reduces fuel consumption.
- Prevents bin overflow.
- Optimizes collection routes.
- Improves city cleanliness.

Applications

- Smart cities
- Municipal waste management
- Recycling systems
- IoT-based garbage monitoring

Conclusion

RL improves waste collection efficiency by learning optimal routes and schedules from continuous environmental updates.

Q8. Evaluate how Reinforcement Learning can be used in network bandwidth allocation in telecommunications. Define states, actions, and rewards, and examine how the system adapts to dynamic conditions. (5 Marks)

Aim

To optimize network bandwidth allocation using Reinforcement Learning.

Theory

RL dynamically allocates bandwidth according to network traffic and user demand. It continuously learns the best allocation strategy to improve network performance.

MDP Components

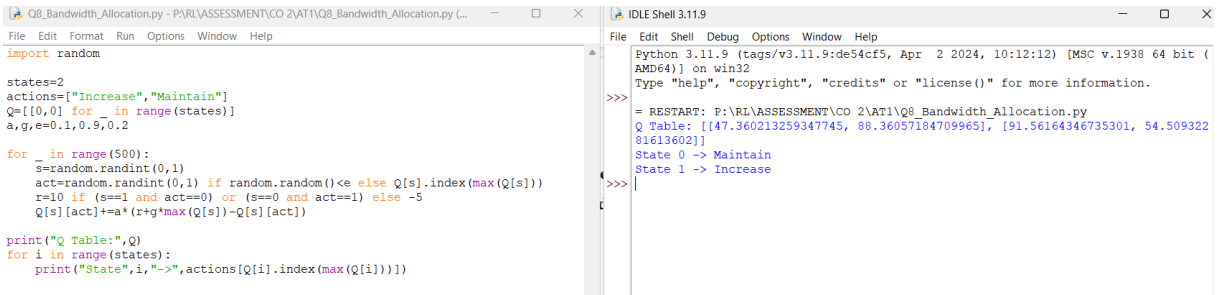
States: Low Traffic, High Traffic

Actions: Increase Bandwidth, Maintain Bandwidth

Rewards: +10 for efficient allocation, -5 for congestion

Policy: Select the best bandwidth allocation.

Python Code



Code Explanation

Step 1: Define two network states.

- State 0 → Low Traffic
- State 1 → High Traffic

Step 2: Define two actions.

- Increase Bandwidth
- Maintain Bandwidth

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Select an action using the ε-greedy strategy.

Step 6: Give rewards.

- Efficient bandwidth allocation → **+10**
- Poor allocation → **-5**

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Display the optimal bandwidth allocation policy.

Advantages

- Reduces network congestion.
- Improves bandwidth utilization.
- Adapts to changing traffic.
- Enhances user experience.

Applications

- Mobile networks
- Wi-Fi systems
- Internet service providers
- Cloud computing

Conclusion

RL enables efficient bandwidth allocation by adapting to changing network conditions in real time.

Q9. Analyze the application of Reinforcement Learning in portfolio management in finance. Discuss advantages over traditional models and challenges such as risk and delayed rewards. (5 Marks)

Aim

To optimize investment decisions using Reinforcement Learning.

Theory

RL assists investors by learning investment strategies from market conditions. It maximizes long-term returns while managing financial risks through continuous learning.

MDP Components

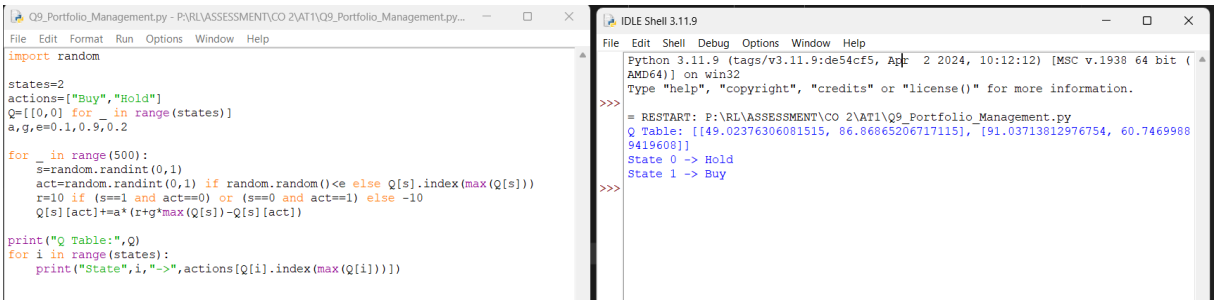
States: Market Up, Market Down

Actions: Buy, Hold

Rewards: +10 for profit, -10 for loss

Policy: Select the action with the highest Q-value.

Python Code



Code Explanation

Step 1: Define two market states.

- State 0 → Market Down
- State 1 → Market Up

Step 2: Define two actions.

- Buy

- Hold

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Select an action using the ε-greedy strategy.

Step 6: Give rewards.

- Profitable investment → **+10**
- Loss-making decision → **-10**

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Display the best investment policy.

Advantages

- Learns market behavior.
- Optimizes investment returns.
- Adapts to market changes.
- Supports automated trading.

Applications

- Stock trading
- Mutual funds
- Investment advisory systems
- Financial analytics

Conclusion

RL provides intelligent portfolio management by learning investment strategies that maximize long-term profit while handling market uncertainty

Q10. Justify the use of Reinforcement Learning in renewable energy management systems. Define the RL framework components and discuss the impact of environmental uncertainty on decision-making.

Aim

To optimize renewable energy usage using Reinforcement Learning.

Theory

RL manages renewable energy resources by deciding when to store, use, or distribute electricity based on production and demand. It adapts to changing weather and energy consumption patterns.

MDP Components

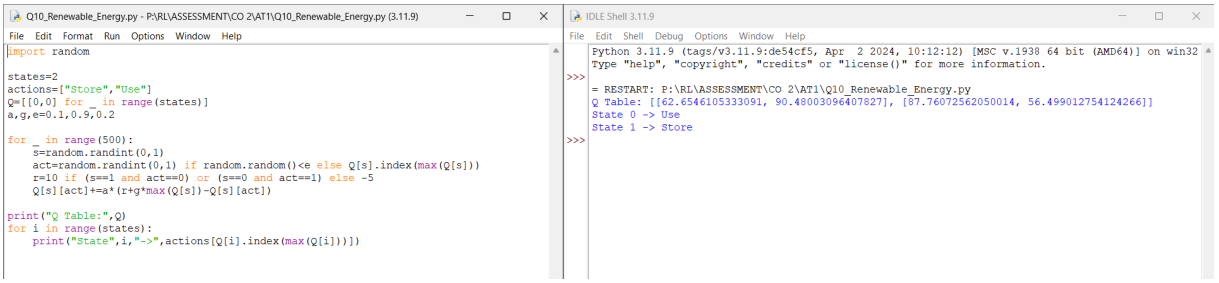
States: Low Energy, High Energy

Actions: Store Energy, Use Energy

Rewards: +10 for efficient utilization, -5 for energy waste

Policy: Select the action with the highest Q-value.

Python Code



Code Explanation

Step 1: Define two energy states.

- State 0 → Low Energy
- State 1 → High Energy

Step 2: Define two actions.

- Store Energy
- Use Energy

Step 3: Initialize the Q-table with zeros.

Step 4: Train the agent for 500 episodes.

Step 5: Select an action using the ε-greedy strategy.

Step 6: Give rewards.

- Efficient energy management → **+10**
- Energy wastage → **-5**

Step 7: Update the Q-table using the Q-Learning formula.

Step 8: Display the optimal energy management policy.

Advantages

- Reduces energy waste.
- Improves battery utilization.
- Adapts to weather changes.
- Supports sustainable energy management.

Applications

- Smart grids
- Solar power systems
- Wind energy farms
- Battery storage systems

Conclusion

Reinforcement Learning improves renewable energy management by learning optimal decisions under uncertain environmental conditions, leading to efficient and sustainable energy utilization.